

PRACTICAL COURSE WORKBOOK

Health Informatics & Digital Health

Work with digital-health information responsibly

LEVEL	Beginner
GUIDED TIME	5 hours across 12 lessons
PROJECT	a safe, fictional health-data workflow and review
SKILLS	Health IT, EHR, FHIR, HL7, Telemedicine, Critical evaluation
EDITORIAL STATUS	Structured draft v0.9

WHAT YOU WILL BE ABLE TO DO

- Use Health IT appropriately in a realistic, bounded task.
- Use EHR appropriately in a realistic, bounded task.
- Use FHIR appropriately in a realistic, bounded task.
- Use HL7 appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners exploring health technology without replacing clinical expertise.

BEFORE YOU BEGIN

- Basic data literacy

This open workbook is a structured draft undergoing course-specific editorial review. It does not provide certification.

COURSE MAP

From foundation to finished work

Health Informatics & Digital Health is a structured, practical course covering Health IT, EHR, FHIR, HL7, Telemedicine. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Health IT: Health context	1. Care pathways with Health IT 2. Health data with EHR 3. Stakeholders with FHIR
02 EHR: Digital systems	1. Standards with HL7 2. Interoperability with Telemedicine 3. Workflows with Health IT
03 FHIR: Quality and safety	1. Validation with EHR 2. Privacy with FHIR 3. Human oversight with HL7
04 HL7: Applied evaluation	1. Implementation with Telemedicine 2. Monitoring with Health IT 3. Equity with EHR

HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Health Informatics & Digital Health project using Health IT, EHR, FHIR.

DELIVERABLE	A working Health Informatics & Digital Health artefact plus an evidence-based self-review.
TOOLS	A suitable Health IT environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Health IT.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This material is educational and does not provide medical advice or replace qualified clinical judgement.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

Global strategy on digital health 2020-2027

World Health Organization - reviewed 2026-08-21
<https://www.who.int/publications/item/9789240116870>

FHIR specification

HL7 International - reviewed 2026-08-21
<https://hl7.org/fhir/>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Health IT scope.